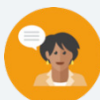


8.5 Surface Area of Rectangular Prisms and Pyramids

Lesson Introduction

 Benchmark	MA.6.GR.2.4 Given a mathematical or real-world context, find the surface area of right rectangular prisms and right rectangular pyramids using the figure's net.		 Learning Targets	- I can find the surface area of a rectangular prism using a net or the figure. - I can find the surface area of a rectangular pyramid using a net or the figure.	
 Benchmark Coherence	Building On		Working Toward		
	N/A		MA.912.GR.4.6 Solve mathematical and real-world problems involving the surface area of three-dimensional figures limited to cylinders, pyramids, prisms, cones and spheres.		
 Essential Questions	- What does the net of a rectangular prism look like? A rectangular pyramid? - What is surface area? - What strategies can you use to find the surface area of a rectangular prism or pyramid? - How can a net be used to find the surface area of a rectangular prism or pyramid?				
 Lesson Narrative	In this lesson, students are introduced to finding the surface area of nets. Through guided instruction, students find the surface area of a rectangular prism and pyramid using a net by first labeling given dimensions and finding the area of each face. By the end of the lesson, students show their learning by identifying all dimensions, labeling nets accurately, and finding the surface area of real-world examples of rectangular prisms and pyramids.				
 Component Summary	8.5.1 Warm-Up (~5 min.)	Which One Doesn't Belong? activity with patterns (<i>SE p. 25</i>)	8.5.4 Guided Instruction (10-15 min.)	Introduction to pyramids and finding the surface area using nets (<i>SE p. 28</i>)	
	8.5.2 Guided Instruction (10-15 min.)	Guided instruction on finding the surface area of rectangular prisms using nets (<i>SE p. 25</i>)	8.5.5 Collaboration (~5 min.)	Collaborate to find the surface area of a pyramid (<i>SE p. 29</i>)	
	8.5.3 Collaboration (~10 min.)	Compare student work (<i>SE p. 27</i>)	8.5.6 Your Turn! (~5 min)	Individual practice determining surface area of rectangular prisms and pyramids within real-world contexts (<i>SE p. 30</i>)	
	8.5.7 Wrap-Up (~5 min.)	Assessment on using a net to label dimensions and find surface area of a rectangular prism (<i>SE p. 31</i>)			
 Resources	Glossary		Materials		Additional Preparation
	<u>Key Terms:</u> - Surface area - Slant height <u>Additional Terms:</u> - Net - Face		(Optional) Four-function calculator (Optional) Graph paper to draw and cut out nets or preprinted nets		Examples of nets of square pyramids and of rectangular pyramids may be beneficial.

 Teacher Tips	Common Misconceptions • Students may miscalculate the missing dimensions. • Students may struggle with correctly labeling the dimensions on the nets. • Students may forget to find the sum of all the areas of the faces or miss one of the faces. • Students may confuse the height and the slant height. • Students may assume that all the triangle faces in a pyramid have the same dimensions or the same area.	Things to Consider • Finding the surface area of three-dimensional figures is a lengthy process for students to execute. Provide different methods to finding the surface area and checkpoints throughout problem-solving to assist students with resilience. Monitor student problem-solving closely for misunderstanding. • Reinforce, as nets and as solids, that all rectangular prisms have exactly six faces and all rectangular pyramids have exactly five faces. Their combined surface area should include the sum of six faces and five faces, respectively. • Determine which problems, if any, you think your students would benefit from using a calculator on. When determining which problems, consider the difficulty of the problem and if that difficulty would be a barrier to student learning.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Collect and Display Consider implementing Collect and Display routines after completing the 8.5.3 Guided Practice. While the students are completing the question, collect comments they make during their discussions about the strategies. Display these comments and others to brainstorm what these comments could be referring to when their peers made them.	MA.K12.MTR.7.1: Real World Contexts This standard emphasizes experiencing math through real-world examples. In this lesson, students explore various authentic examples of rectangular prisms and work with nets to find and label dimensions and find the total surface area in real-world contexts.
 Differentiate Instruction	Consider providing graph paper, rulers, markers, scissors, and tape for students to recreate the nets in 8.5.3 Collaboration and 8.5.6 Your Turn!. By providing these hands-on materials and visuals, students can create prisms to scale and physically label each face as they measure and find the area, additionally helping students link the characteristics of two-dimensional nets to the characteristics of three-dimensional rectangular prisms.	
Lesson Notes:		

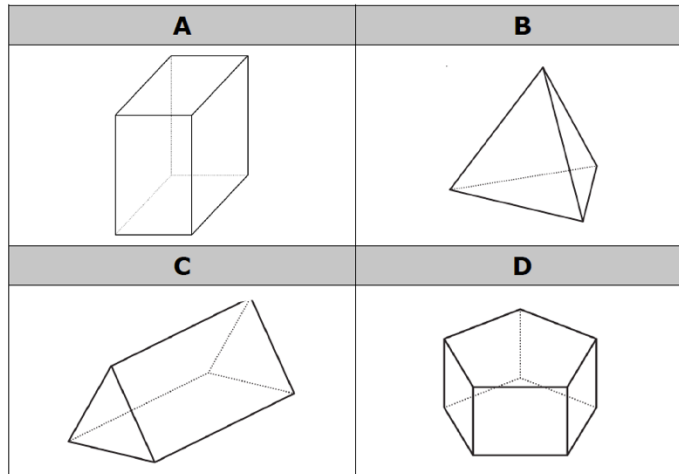
8.5.1 Warm-Up: Which One Doesn't Belong?

Component Overview: Students are presented with four patterns. Students must determine which pattern does not belong with the other three and must justify their selection.



8.5.1 Warm-Up: Which One Doesn't Belong?

1. Four figures are shown.



- a. Which one doesn't belong?

Answers will vary.

- b. Justify your choice using mathematics.

A is the only one where all of the faces are rectangles; A is the only rectangular prism.

B is the only one where all the faces are triangles; or the only one that doesn't include rectangular faces; or the only one that isn't a prism; or the only one that is a pyramid.

C is the only one that include rectangular and triangular faces.

D is the only one with faces that aren't rectangles or triangles; is the only one with pentagonal faces.



Consider a kinesthetic approach to eliciting student responses to this activity. Post a picture of each pattern, one per corner, and ask the students to stand in the corner with the picture that does not belong. Ask students to share their justifications first with their group, then to other groups in other corners. Allow students to walk between corners as an added kinesthetic approach to clarify and justify thinking.

8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

Component Overview: Students are introduced to finding the surface area of two rectangular prisms using nets. Students are given a net on a grid and must find all the dimensions, the area of each face, and the surface area. Students are also given a real-world example of a rectangular prism and must draw the net, label it, and find the surface area. Through guided instruction and discussion, position students to identify the shape of each face and to find the area of each shape.

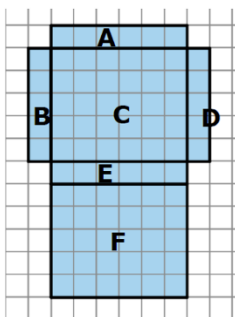


8.5.2 Guided Instruction: Surface Area of a Rectangular Prism

To find the surface area of a three-dimensional object, use the nets of a shape to find the area of each face.

Surface area is the total area of the two-dimensional surfaces that make up a three-dimensional object

1. A net of a rectangular prism is shown below with each face labeled.
 - a. Complete the table to find the area of each face. Each unit of the grid is 1 centimeter (cm).



Face	Dimensions	Area
A	1 cm x 6 cm	6 cm ²
B	1 cm x 5 cm	5 cm ²
C	5 cm x 6 cm	30 cm ²
D	1 cm x 5 cm	5 cm ²
E	1 cm x 6 cm	6 cm ²
F	5 cm x 6 cm	30 cm ²

- b. How can the net and the area of the six faces be used to find the surface area of the prism?
You can add up the areas of all the sides to find the surface area.
 - c. What is the surface area of this rectangular prism?
82 cm²



When calculating surface area, find the sum of the areas of each face of the figure. Use the figure's net to label the dimensions needed to calculate the area of each face.



For question 1, elicit student thinking on key elements about the figure before completing the table by asking:

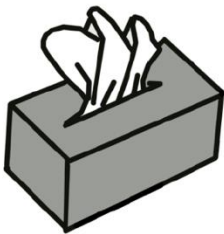
- How can we find the dimensions of each shape?
- How can we verify that the net of the prism is made up of rectangles and not squares?
- Since the net is on a grid, what is another method we could use to find the surface area instead of adding all the areas?



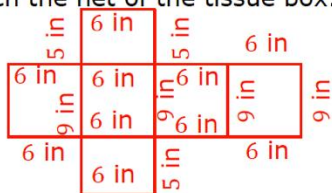
Before reading the blue box explaining how to calculate surface area aloud, have students read it individually to themselves, circling any words they do not recognize or are unfamiliar with. Then define any needed words and read the description aloud. This auditory approach may be especially helpful for ELL students by frontloading key vocabulary content that is needed to access procedural knowledge on finding surface area using a net.

8.5.2 Guided Instruction: Surface Area of a Rectangular Prism (continued)

2. A tissue box is 9 inches (in.) long, 5 in. wide, and 6 in. tall.



- a. Sketch the net of the tissue box.



- b. Label the dimensions of each face on the net.
- c. Find the area of each face. Include units.
 45 in^2
 45 in^2
 54 in^2
 54 in^2
 30 in^2
 30 in^2
- d. Add the area of the faces to find the surface area of the tissue box. Include units.
 258 in^2



Collect and Display

After students complete question 2, prompt students to consider a different arrangement of the faces of the tissue box. Consider displaying labeled nets that are not identical to each other but still use the correct dimensions, then discuss how they are all correct. Students may point out that all the nets have six shapes. Students may also share that when folded, the nets form a rectangular prism. Point out that the arrangement of the six shapes could be different, but when folded, they still create the same rectangular prism.

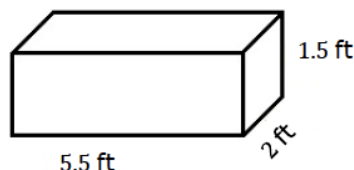
8.5.3 Collaboration: Analyzing Student Work

Component Overview: Students work in partners to solve real-world problems related to the surface area of rectangular prisms, including labeling a net's dimensions and finding the surface area by first finding the area of each face. Consider regrouping and sharing ideas at the end of 8.5.3 Collaboration section as students compare two different strategies used to find surface area.



8.5.3 Collaboration: Analyzing Student Work

- Jillian and Victor are finding the surface area, in square feet (ft^2), of the rectangular prism shown and each got the same answer; however, they solved the problem differently.



Jillian's Work	Victor's Work
Side 1: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	2 sides: $2(5.5 \text{ ft} \times 1.5 \text{ ft}) = 16.5 \text{ ft}^2$
Side 2: $5.5 \text{ ft} \times 1.5 \text{ ft} = 8.25 \text{ ft}^2$	2 sides: $2(5.5 \text{ ft} \times 2 \text{ ft}) = 22 \text{ ft}^2$
Side 3: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	2 sides: $2(2 \text{ ft} \times 1.5 \text{ ft}) = 6 \text{ ft}^2$
Side 4: $5.5 \text{ ft} \times 2 \text{ ft} = 11 \text{ ft}^2$	Total Surface Area: 44.5 ft^2
Side 5: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Side 6: $2 \text{ ft} \times 1.5 \text{ ft} = 3 \text{ ft}^2$	
Total Surface Area: 44.5 ft^2	

Discuss with your partner each student's strategy and how they are similar or different. Draw arrows to show connections between each student's work.

Answers will vary.

Jillian found the area of each side then add them all together. Victor knew that 2 sides that were parallel to each other would have the same measurements therefore, he multiplied the 3 different side areas by 2 and then added. They both added the side areas.



Extend and connect geometric concepts by prompting students to identify congruent shapes. Ask students how to determine if a shape is similar or congruent, then connect this concept to nets by identifying each pair of congruent sides and justify their reasoning. Students may respond that the top and the bottom are congruent because the dimensions are identical.



Collect and Display

After students have discussed each student's strategy and how they are all similar and different, consider having students write which strategy they prefer on a sticky note. Display these sticky notes on a shared classroom space and discuss which strategy is more popular. Point out that both strategies work, but the process and approach differ. Encourage students to try the strategy they did not pick on a future problem and return to this discussion after question 1 in 8.5.6 Your Turn!.

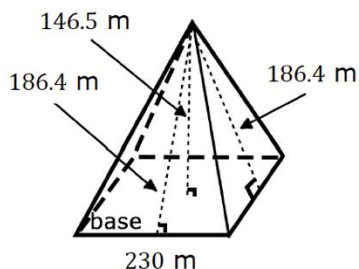
8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

Component Overview: Students are introduced to pyramids using Egyptian pyramids and focus on finding the surface area when given a 3D shape and/or net. Instruction should include the difference between the height and the slant height. Students label the net, find the area of each face, and find the total surface area, then select correct options to complete statements on finding the surface area of pyramids.



8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid

1. Around 2500 B.C. the Egyptians began building pyramids with intricate hallways and layouts to house the tombs of pharaohs and their families. One of the most famous is the Pyramid of Khufu, seen below. The Egyptians designed the pyramid to have a square base with approximate side lengths of 230 meters, a height of 146.5 meters, and a slant height of 186.4 meters.



Looking at the model of the pyramid, why is there a measurement called height and a measurement called slant height?

The height is straight up the middle of the pyramid to the highest point. The slant height is up the side of the pyramid to the top because the side is slanted.

The slant height of the pyramid represents the height of each triangular face.



When finding the surface area of a pyramid, use the **slant height** to find the area of the faces.



While looking at the model of the pyramid in question 1, prompt students to consider the key elements distinguishing the height from the slant height by asking:

- Where in the diagram are the height of the pyramid and the slant heights of the faces located?
- Why does the diagram include more than one slant height and only one height?
- What does the square box at the bottom of the height indicate?

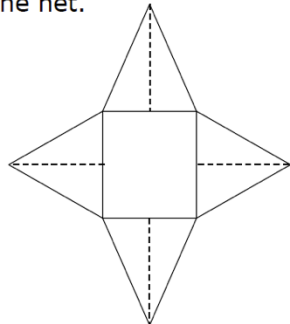


Consider using a hands-on demonstration to help students differentiate between slant height and pyramid height. Use a physical net of a pyramid and ask students to identify and point at the slant height of each triangular face on the physical net and on the drawn net in question 1. Then fold the net together and ask students to identify the pyramid height on the physical net and on the drawn pyramid.

Students may find it easier to understand slant height by comparing it to the stairs needed to walk up the side of the pyramid.

8.5.4 Guided Instruction: Surface Area of a Rectangular Pyramid (continued)

2. Using the information about the pyramid, complete the table to determine the surface area of the pyramid. Begin by labeling the sides of the net.



Encourage students to analyze the net provided prior to completing the table. Students may point out that the center shape is a square because all the side lengths are the same. Students may also note that all the faces attached to the square are triangles. Students may also notice that squares are special kinds of rectangles.

Also compare the net of a pyramid to the net of a rectangular prism and look for similarities and differences, like the number and shape of the faces.

Side	Formula	Area
Rectangular base	$A = l \cdot w$	$A = 230 \cdot 230$ $A = 52,900 \text{ m}^2$
Triangular Face 1	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 2	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 3	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
Triangular Face 4	$A = \frac{1}{2} \cdot b \cdot h$	$A = \frac{1}{2} \cdot 230 \cdot 186.4$ $A = 21,436 \text{ m}^2$
		Surface Area: $138,644 \text{ m}^2$

3. Complete the following statement.

To find the surface area of a rectangular pyramid you need

the sum of the ☐ height ☒ area of the base plus the area of

☐ one of ☒ all the other faces.

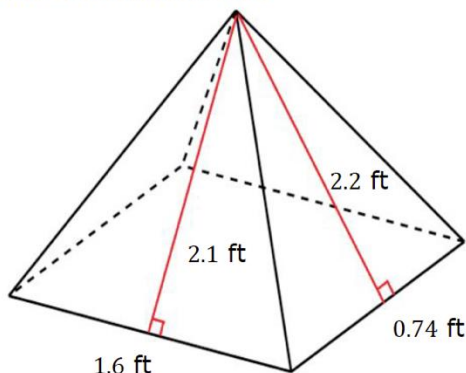
8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

Component Overview: Students work with a partner to find the total surface area of a labeled diagram of a real-world example of a rectangular pyramid.



8.5.5 Collaboration: Surface Area of a Rectangular Pyramid

Papier-mache is a type of art that uses paper and glue. Jose is using the art of papier-mache to create a rectangular pyramid for his art project. To create his project, Jose needs to determine the amount of area that will be covered in newspaper. Below is the drawing Jose created to find the surface area.



1. Work with your partner to find the surface area of the pyramid.

$$A_{\text{rectangular face}} = 1.6 \times 0.74 = 1.184 \text{ ft}^2$$

$$A_{\text{triangular face 1}} = \frac{0.74 \times 2.2}{2} = 0.814 \text{ ft}^2$$

$$A_{\text{triangular face 2}} = \frac{1.6 \times 2.1}{2} = 1.68 \text{ ft}^2$$

$$A_{\text{triangular face 3}} = \frac{0.74 \times 2.2}{2} = 0.814 \text{ ft}^2$$

$$A_{\text{triangular face 4}} = \frac{1.6 \times 2.1}{2} = 1.68 \text{ ft}^2$$

$$\begin{aligned} \text{Total Surface Area} &= 1.184 + 0.814 + 1.68 + 0.814 + 1.68 \\ &= 6.172 \text{ ft}^2 \end{aligned}$$



To scaffold students through the process to find the surface area, pose these questions:

- What are the five shapes that make up this pyramid?
- Which of these shapes are congruent?
- What are the dimensions of the base and of each face?
- How can the area of the shapes be found?
- Once the areas are found, how can the total surface area be found?



Finding the surface area of three-dimensional figures is a process that involves many steps for students to execute. Encourage students to draw labeled pictures as they go through the steps to find the surface area. Also encourage students to use a table, like the table included in the 8.5.4 Guided Instruction, for them to organize their calculations.



Consider an interdisciplinary lesson with an art teacher using papier-mâché to conceptualize and represent surface area. Have students create their own version of a net of a pyramid using cardstock paper, then measure the surface area to determine how much newspaper they would need to cover the pyramid with three layers. If time allows, let students create these pyramids and papier-mâché the outside surface area to display in the classroom.

8.5.6 Your Turn!

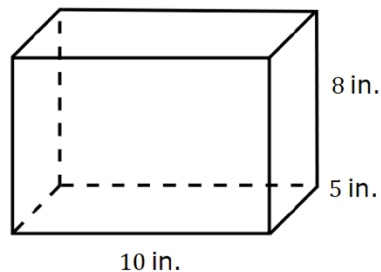
Component Overview: An image of a rectangular prism is given, and students practice finding the surface area. Students are also given a real-world scenario involving a rectangular prism and must find the surface area. While some students work individually to practice these skills, consider working with a group of students who may need additional support identifying the shape of each face, labeling the dimensions, and finding the area of each shape. Allow students to draw nets that are not identical to each other.



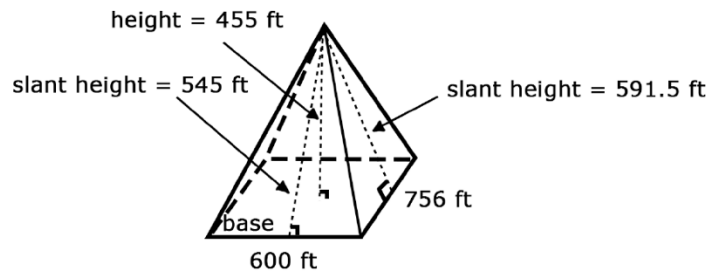
8.5.6 Your Turn!

1. Darsey wants to build trinket boxes made of wood to sell at the local craft festival. A drawing of her box is shown. How much wood does Darsey need to buy for each trinket box?

340 in^2



2. Find the surface area of a rectangular pyramid that has a rectangular base with a side length of 600 feet (ft), which is the base of the triangle with a slant height of 545 ft, and a side length of 756 ft, which is the base of the triangle with a slant height of 591.5 ft.



$\text{Surface Area} = 1,227,774 \text{ ft}^2$



Consider prompting students to draw a net for question 1, then recording the area of each shape inside of the shape on their drawings of the net. By labeling the area of each face, students practice a visual strategy that will assist them with finding the surface area.



For question 1, consider challenging students by presenting the same labeled rectangular prism but turned differently, such as one prism shown horizontally and the other prism shown vertically. Ask half of the class to solve the problem using the horizontal figure and the other half of the class to solve the problem using the vertical figure. Providing the same image in two different views reinforces that surface area can be found regardless of the positioning of the rectangular prism.



Prior to completing question 2, prompt students to notice specific characteristics of the pyramid. Students may recognize that the height of the pyramid is not necessary to calculate the surface area. Students may also see that since two different slant heights are labeled, all the triangular faces are not identical.

8.5.7 Wrap-Up: Ticket Out the Door

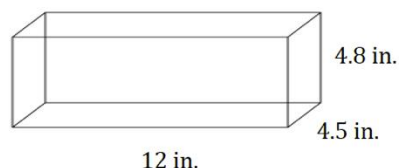
Component Overview: Students individually solve two problems by labeling a net and finding surface area within a real-world context. Consider using student responses as a formative assessment of this lesson's learning targets.



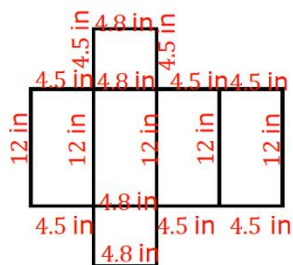
8.5.7 Wrap-Up: Ticket Out the Door

Ted is preparing a holiday display for the post office. He takes an unfolded box and uses it to determine the amount of wrapping paper needed to wrap the box.

The shipping box Ted uses for the display is shown.



- Write the measurements on the net of the box.



- Use the net to find the surface area of the shipping box. Include units.
 266.4 in^2



Post these questions to reinforce the essential questions for early finishers or all students based on available time:

- How do you know what the shapes of the faces are?
- Which faces have the exact same area? How can you tell? How many faces have the same area?
- How can you use the net to find the total area of all the surfaces of this prism?